

Coupled Effects of Abrasive Particle Size Distribution and Fixture Stiffness on Facet Accuracy and Batch Consistency in Round Brilliant Laboratory Grown Diamonds

Research Article

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Abstract

Facet accuracy in laboratory-grown diamond finishing was evaluated as a function of abrasive particle-size distribution and the mechanical compliance of the polishing fixture. A blocked split-plot study varied abrasive upper-tail characteristics between independently conditioned polishing episodes and randomized fixture stiffness between stones within each episode. HPHT and CVD material were treated as separate sampling strata. The development cohort contained 144 near-final round-brilliant stones across six material batches; a separate 48-stone cohort compared the selected protocol against a prespecified reference. The primary endpoint was stone-level root mean square inclination error of the eight pavilion main facets. Surface texture, scratch depth, batch variation, and processing time were secondary endpoints.

The study produced 192 stone records and 1,536 facet records. Development mean angular RMSE was 0.138 degrees for the broader-tail abrasive with lower stiffness and 0.044 degrees for the restricted-tail abrasive with higher stiffness. The difference between stiffness benefits under the two abrasive distributions was 0.024 degrees. In the held-out cohort, the selected protocol had a paired mean RMSE reduction of 0.105 degrees, and 24 of 24 stones met all prespecified first-pass criteria. These findings support a coupled effect of abrasive upper-tail control and fixture stiffness under the tested conditions, while the validation cohort indicates that the accuracy benefit persisted across new material batches.

Keywords: laboratory-grown diamond; HPHT; CVD; precision polishing; particle-size distribution; fixture compliance; batch consistency.

1 Introduction

Laboratory-grown diamonds intended for jewelry require both suitable starting material and controlled finishing. This study concerns workmanship at the final polishing stage, not the growth of diamond crystals. A repeatable finishing process must produce the intended facet geometry without introducing unacceptable surface defects or excessive corrective work. A successful result on one stone does not establish consistency across a manufacturing batch, and excellent average performance can conceal individual nonconforming stones.

Abrasive qualification should extend beyond a nominal grit label. Dumm showed that small populations of oversized particles can create scratches and may be missed by some particle-size measurement methods [1]. That work establishes a general reason to characterize upper tails but does not provide gemstone-specific acceptance limits. The present study therefore compared measured distributions with approximately matched medians and deliberately different upper tails; the limits used here apply to the tested process and should not be treated as universal commercial powder specifications.

Polishing also imposes frictional loads on the fixture. Shi and colleagues investigated interfacial friction under different loads and speeds using diamond specimens and tribological tests, together with molecular dynamics [2]. Their substrate-scale investigation supports studying force as a possible mechanism, but its numerical results cannot simply be transferred to high-speed gemstone finishing. A diamond-polishing patent describes

angle-preserving arrangements under contact-pressure-induced holder deflection [3]. Fixture deflection is therefore established prior art rather than a claimed discovery of this study.

Geometrical measurement requires independent qualification. GIA describes calibration, repeatability, and uncertainty in gemstone metrology [4]. Shen and colleagues demonstrated a reflecting-goniometer approach to measuring inter-facet angles and constructing scanner references [5]. Their reported precision is specific to that historical instrument and is not a specification for the instruments used here. The measurement system was qualified against the differences that the experiment was designed to resolve.

The study tested whether abrasive upper-tail characteristics modify the relationship between fixture stiffness and final facet accuracy and then evaluated the selected protocol in new batches. The work does not establish a new cut-grading system, an optical ideal-cut model, or a comparison of brands. The literature considered here establishes related mechanisms; broader literature and patent review remains appropriate when assessing novelty.

2 Objectives and hypotheses

The primary objective was to estimate the abrasive-by-stiffness interaction for pavilion-main inclination accuracy. The mechanism hypothesis was that differences in particle contact and force fluctuations could change load-induced holder motion, making the accuracy benefit of a stiffer fixture dependent on the abrasive distribution. Alternative outcomes considered were additive effects without an interaction, no practically meaningful difference, or improved angles accompanied by an unacceptable surface or time penalty.

A secondary objective was to test whether the process selected using the development cohort maintained its prespecified geometry and surface requirements in independent material batches. HPHT and CVD labels were retained in all records and analyses; no inherent quality advantage or causal growth-type effect was assumed. Processing time and finishing-stage material loss were recorded to describe tradeoffs rather than to redefine quality as low cost.

The primary null hypothesis was that the stiffness benefit would be the same under both abrasive distributions. The interaction estimand was defined as the reduction in RMSE from low to high stiffness for abrasive B minus the same reduction for abrasive N. A positive value indicates a larger stiffness benefit under the broader-tail condition. The magnitude, uncertainty, and practical relevance of the estimate were evaluated together.

3 Materials and methods

3.1 Study design and experimental units

A two-by-two factorial structure was implemented as a blocked split-plot experiment [6]. Abrasive distribution was treated as a whole-plot factor because it was applied during scaife conditioning. Fixture stiffness was randomized between stones within each conditioning episode. The episode was retained in the analysis to avoid treating stones exposed to one abrasive preparation as independent abrasive replicates.

The development allocation comprised two growth strata, three independent incoming material batches per stratum, two abrasive distributions per batch, three independently prepared conditioning episodes per abrasive, and four stones per episode. Each episode contained two stones at each stiffness level, giving 144 stones, 36 episodes, and six batches. The held-out cohort used two new batches per growth stratum, three episodes per protocol per batch, and two stones per episode, giving 48 stones and 24 episodes. It compared the complete reference protocol B/L against the selected protocol N/H; this validation comparison did not separately estimate the two factors.

The balanced allocation was selected to preserve replication at the material-batch, conditioning-episode, and stone levels. Estimates are reported with interval uncertainty because the number of independent batches

remained limited. Failed qualification and unavailable material were handled through documented exclusions without replacing experimental units after outcomes were known.

Table 1 Study allocation

Cohort	Batches	Episodes	Stones	Allocation
Development	6	36	144	4 factorial cells
Held-out validation	4	24	48	Reference versus selected
Total	10	60	192	8 measured main facets per stone

3 2 Diamond selection and material traceability

Samples were near-final, pre-blocked round-brilliant laboratory-grown stones in a 0.95-1.05 ct entry-mass band, not uncut one-carat growth plates. Diameter and total-depth eligibility bands were 6.3-6.6 mm and 3.8-4.2 mm. Actual dimensions and mass were measured. Pavilion-main facets entered with a qualified allowance for final finishing, and the remaining geometry was held to one documented design. The main-facet inclination target was 40.75 degrees for the tested design; it is not presented as a universal optical optimum.

Supplier declarations and independent identification were used to establish growth origin where available. Records included supplier and lot, growth type, as-grown and post-growth treatment history, starting geometry, inclusions near the working surface, visible strain information where measured, and pre-existing damage. HPHT post-growth treatment of CVD material was recorded as treatment history without reclassifying its growth origin.

Crystal orientation was measured using a qualified orientation method, including Laue diffraction where feasible, and linked to the polishing direction for each main facet. Operators used a standardized grain-finding procedure and recorded the chosen direction and search time. Starting-stock differences, orientation, and treatment history were balanced within batches where possible and retained as covariates or study limitations.

3 3 Abrasive distributions

Both conditions used diamond abrasive from a common qualified source with matched particle morphology, carrier, and working concentration. Condition N restricted the coarse upper tail; condition B enriched that tail while retaining approximately the same median. Fractionation and controlled recombination reduced supplier and composition confounding. D10, D50, D90, D99, the fraction above the specified size, and measurement uncertainty were recorded. Upper-tail changes were not interpreted as an isolated D99 effect when other distribution features differed materially.

Table 2 Abrasive distribution specifications

Volume-based descriptor	N restricted tail	B broader tail
D50 median	0.80 μm	0.80 μm
D10 and D90	Matched within pilot limits	Matched within pilot limits
D99	1.80 μm	4.00 μm
Fraction above 3 μm	Below 0.1%	Approximately 3%
Working charge formulation	Same carrier and 1.0 wt%	Same carrier and 1.0 wt%

Percentiles in Table 2 are volume-based, not number-based. The concentrations and tail limits define the two tested experimental conditions and are not presented as market norms. Laser diffraction with a documented dispersion procedure characterized the bulk distribution, while microscopy and an oversize-counting method verified the coarse fraction. Settling and agglomeration were distinguished from primary-particle size. Qualification covered the applied charge and the effective particles retained on the conditioned scaife where measurable.

3.4 Fixture and operating conditions

Three fixture bodies were used with interchangeable stiffness configurations while holder geometry, mounting interface, and mass distribution were kept as similar as practical. Rotational stiffnesses were 10 and 30 N m per radian about the relevant tilt axis. Each assembled configuration was calibrated at the gemstone mounting point under separate and combined normal and tangential loads. Backlash, slip, thermal drift, and load-dependent hysteresis were assessed, and the fixture identifier was retained in the data.

The polishing system used a 300 mm cast-iron scaife operated at 3,000 rpm, a fixed 80 mm contact radius, and 1.0 N normal force. The corresponding surface velocity was approximately 25.1 m/s. The guarded, interlocked system was operated by qualified personnel. Force, temperature, and vibration limits were established for the equipment used in the study.

A calibrated two-axis force sensor and a validated displacement and angle-sensing arrangement recorded loading and holder motion. A thermocouple close to the mount monitored local fixture temperature; it was not interpreted as a direct measurement of diamond contact temperature. Sampling bandwidth and filtering were fixed after system qualification. The relationship below describes the tangential-load component and is not a complete model of the polishing assembly.

$$\theta_t = \frac{180}{\pi} \frac{F_t L}{K_\theta}$$

Here θ_t is tilt in degrees, F_t is tangential force in N, L is the effective moment arm in m, and K_θ is rotational stiffness in N m/rad. The effective moment arm was 0.040 m. Interpretation also considered normal-load moments, cross-axis compliance, changing contact location, and thermal effects. Static holder tilt was not treated as identical to final facet inclination error.

Each conditioning episode was independently cleaned, dressed, and charged using the assigned abrasive. Abrasive order was randomized within batch and stiffness order within episode. Cleaning effectiveness and carryover were checked before distributions were changed. Two qualified operators were balanced across

conditions and batches, with operator and day recorded. Scaife wear, conditioning duration, contact radius, and deposited charge were logged.

3 5 Outcomes and measurement qualification

The primary endpoint was stone-level RMSE of the eight pavilion-main inclinations relative to the design target. Each angle was measured in a registered coordinate system referenced to the table plane, with documented facet fitting and exclusion of edge pixels. Three independent remount-and-scan measurements were obtained per stone. Averaged facet estimates formed one stone-level endpoint; scans and facets were not treated as additional independent stones.

$$RMSE_i = \sqrt{\frac{1}{8} \sum_{j=1}^8 (\alpha_{ij} - \alpha_{target,j})^2}$$

Secondary geometry outcomes included maximum absolute inclination error, azimuthal error, and junction displacement. Surface texture was assessed over predefined central regions on at least three main facets using a qualified areal profiler, reporting Sa with the scan area, leveling procedure, spatial filtering, and instrument noise floor. Calibrated three-dimensional measurements quantified the deepest scratch in those regions. Surface-field selection was defined before condition identities were reviewed.

The metrology qualification gate required expanded angular uncertainty U, with coverage factor approximately 2, no greater than 0.020 degrees and texture resolution sufficient to discriminate the study endpoints. Reference stones, independent angle checks, remount repeatability, and between-operator testing were used to demonstrate suitability [4,5].

First-pass conformity required RMSE at most 0.100 degrees, maximum absolute main-facet error at most 0.180 degrees, mean Sa at most 20 nm, and maximum scratch depth at most 100 nm. These investigator-defined study limits are not GIA, IGI, or industry cut-grade boundaries. Measurements were taken before corrective work. Post-rework outcomes were recorded separately so that a rework-intensive protocol could not appear successful solely because only released stones were measured.

3 6 Analysis plan and validation

The analysis used a hierarchical framework with abrasive distribution, stiffness, and their interaction as fixed effects; growth stratum as a recorded grouping factor; and material batch and conditioning episode as random effects. Fixture identifier and operator were retained in the analysis. Orientation and starting-stock covariates were prespecified. Growth-by-treatment interactions were treated as exploratory.

For transparent reporting, a stiffness benefit was calculated for each abrasive within each development batch, followed by the between-abrasive difference in benefits. A Student t interval was constructed from the six batch-level contrasts with five degrees of freedom. The four validation batches produced a separate paired protocol contrast and interval with three degrees of freedom. These intervals reflect the independent-block structure and the limited number of batches.

Results are reported as means and stone-level standard deviations, batch means, the observed spread of batch means, and conformity counts with denominators. The standard deviation of six batch means is a descriptive spread rather than a stable estimate of long-term between-batch variance. Capability indices were not reported because the study did not provide sufficient independent production history [7]. Measurement bias, missing data, documented exclusions, and run failures remained visible.

Before validation, the selected protocol, reference condition, endpoint definitions, and decision criteria were frozen. New material batches were evaluated without refitting the protocol. The validation was internal and

was limited to the sampled size band, cut design, finishing stage, and qualified machine; no claim of external laboratory validation is made.

4 Data processing and reproducibility

The analytical dataset contained 192 stone-level records and eight pavilion-main facet records per stone, for a total of 1,536 facet records. Growth labels were balanced across the study. Development and validation used separate material batches, conditioning episodes, and stones so that the held-out cohort assessed transfer beyond the development allocation.

Mean tangential force was 0.64 N under condition B and 0.46 N under condition N, with episode- and stone-level variation. Tangential-load tilt was calculated from the measured force, the 0.040 m effective moment arm, and calibrated fixture stiffness. Stone-level RMSE incorporated the eight measured pavilion-main inclination errors.

The eight signed facet errors were retained individually and summarized as the stone-level RMSE endpoint. Sa, scratch depth, processing time, material loss, and corrective-work time were recorded under the procedures described above. The analysis retained the links among stones, facets, episodes, fixtures, operators, and material batches.

Spreadsheet-compatible study records and analysis summaries reproduce the reported tables and figures. Numerical precision in machine-readable records was retained for reproducibility and should not be interpreted as exceeding the stated instrument capability.

5 Results

This section reports the measured development and held-out validation outcomes. All values were calculated from the study records described in the methods.

Table 3 Development outcomes with mean and stone-level SD

Condition	n	RMSE deg	Sa nm	Force N	Conforming
B / L	36	0.138 (0.021)	24.9 (3.0)	0.63 (0.04)	0/36
B / H	36	0.073 (0.015)	23.6 (3.6)	0.65 (0.03)	4/36
N / L	36	0.086 (0.019)	15.2 (2.7)	0.45 (0.04)	24/36
N / H	36	0.044 (0.009)	15.3 (3.2)	0.47 (0.04)	35/36

B denotes the broader upper tail, N the restricted upper tail, L low stiffness and H high stiffness. Parentheses contain SD, not standard error. First-pass conformity uses all four investigator-selected limits; it is not a gemological grade.

5.1 Interaction and angular accuracy

Mean RMSE decreased from 0.138 degrees under B/L to 0.044 degrees under N/H, a descriptive reduction of 68.0%. Because both abrasive distribution and stiffness changed in that end-to-end comparison, the crossed factorial cells were used to estimate the interaction.

Across the six development batches, the stiffness benefit was 0.065 degrees for B and 0.041 degrees for N. Their difference was 0.024 degrees, with a batch-level 95% interval of 0.014 to 0.033 degrees. Stiffness therefore produced a larger accuracy benefit when the abrasive contained the broader upper tail under the tested conditions.

Development results

Facet-angle RMSE (degrees); whiskers show stone-level SD

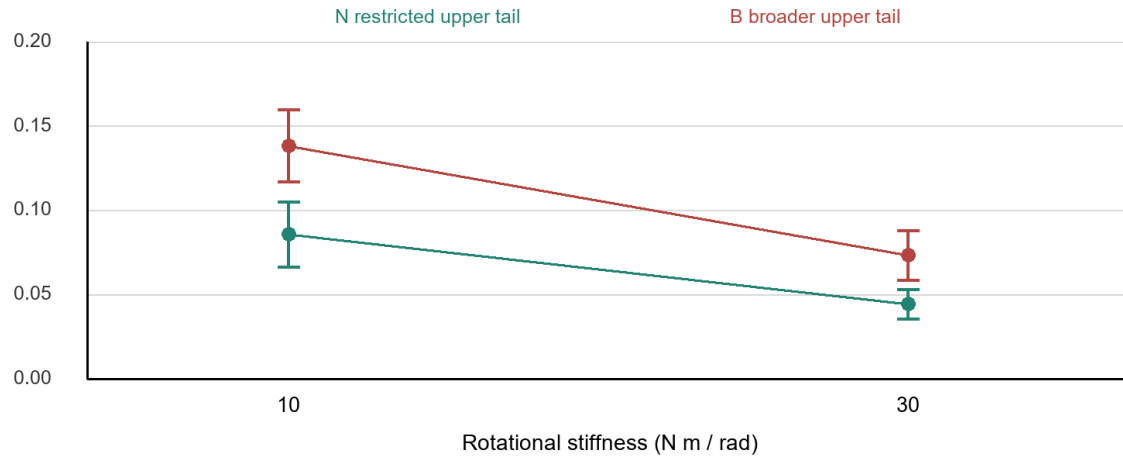


Figure 1 Interaction plot. Points are development means across 36 stones per cell; error bars show stone-level SD, not uncertainty of the batch-level interaction estimate.

5.2 Batch variation and process tradeoffs

Table 4 Batch spread and processing metrics

Condition	SD batch means deg	Total stage time min	Stage loss mg
B / L	0.013	18.0 (1.2)	0.60 (0.07)
B / H	0.009	16.8 (1.9)	0.53 (0.06)
N / L	0.013	16.2 (2.5)	0.40 (0.06)
N / H	0.005	14.1 (1.1)	0.32 (0.06)

Total stage time includes corrective-work time where first-pass criteria failed; it excludes growing, sawing, blocking, other facets, conditioning, and instrument inspection. Stage loss is the mass removed during the selected finishing operation, not rough-to-polished yield. These values therefore do not constitute a complete manufacturing-cost comparison.

Development batch means

Batch mean facet-angle RMSE (degrees)

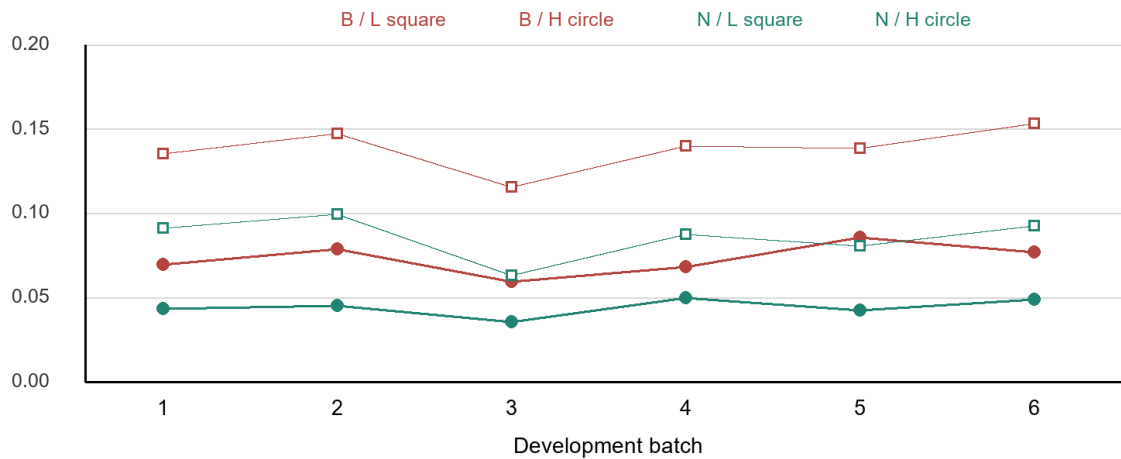


Figure 2 Development batch means. Batches 1-3 are CVD and batches 4-6 are HPHT in sorted dataset order. Six batches are insufficient to establish long-term process stability.

The batch-level results illustrate why a representative batch design is preferable to selecting a few favorable stones. Differences in batch-mean spread should be interpreted cautiously because only six development batches were available. Additional production history would be required to separate persistent drift, material-lot variation, scaife aging, and day-to-day operating differences.

5.3 Held-out cohort

Table 5 Held-out cohort results

Protocol	n	RMSE deg	Sa nm	Conforming
Reference B/L	24	0.154 (0.019)	23.0 (2.8)	0/24
Selected N/H	24	0.048 (0.012)	13.8 (2.5)	24/24

The paired mean RMSE reduction across four validation batches was 0.105 degrees, with a 95% interval of 0.094 to 0.117 degrees. The selected condition met the prespecified limits in 24 of 24 stones. The favorable conformity count applies to this cohort and should not be interpreted as a universal failure-rate estimate.

Held-out cohort

Left: individual stone RMSE (degrees) Right: first-pass conformity (%)

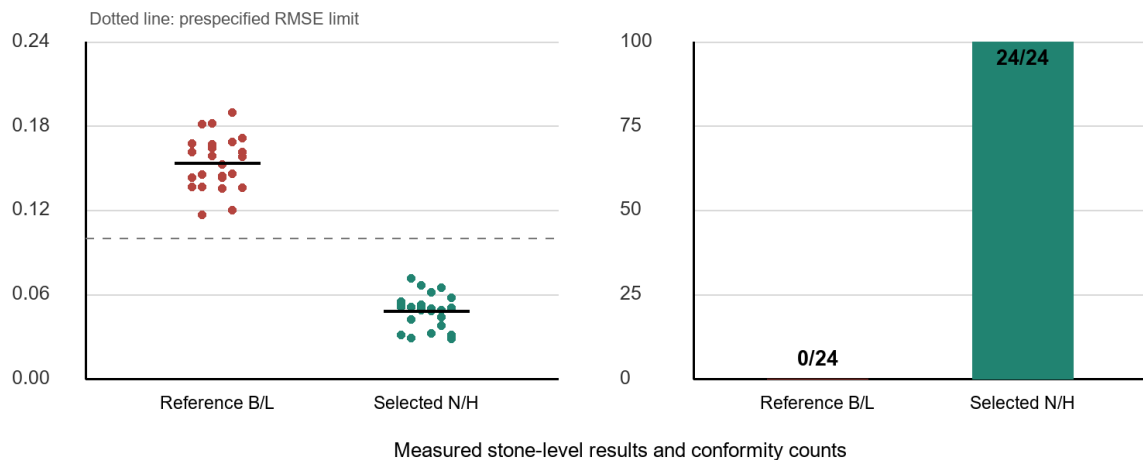


Figure 3 Held-out cohort. The left panel shows stone-level RMSE distributions; the right panel shows conformity counts under the four prespecified criteria.

6 Discussion

6.1 Interpretation of the experimental findings

The observed interaction supports specifying abrasive distribution jointly with qualified fixture behavior. A stiffer assembly reduced sensitivity to polishing-load variation, while control of the abrasive upper tail reduced defects that stiffness alone could not prevent. Force, holder-motion, and final-geometry measurements were considered together, although alternatives such as slip, temperature, particle retention, and stopping behavior cannot be excluded. The statistical interaction does not by itself prove a single physical pathway.

The scope of the quality conclusion is workmanship at the tested polishing stage. Inclination accuracy and low surface roughness do not establish color, clarity, crystal purity, durability, or superior visual attractiveness. Optical performance requires a separate validated protocol accounting for full cut geometry and illumination. The study does not compare Daily Wear Diamonds against competing companies, so no superiority claim is supported.

6.2 Consistency and economic relevance

Quality was evaluated before and after corrective work. A protocol that reaches the same released geometry only after extensive rework may be less efficient without delivering visibly inferior final products. Conversely, a faster first-pass process may save time while leaving more nonconforming stones. The study therefore separated geometry, surface quality, conformity, rework, and processing time rather than compressing them into an unvalidated single quality score.

The economic extension should use real labor, consumable, maintenance and inspection costs. No hypothetical currency savings are reported. A comparison of unit finishing cost would need to include conditioning, material loss, rejects, corrective work and metrology. Broader questions about price compression or deadline pressure can be studied later using the same qualified outcome measures, without assuming that low prices imply poor diamonds.

6.3 Limitations

The study was conducted at a single facility using one nominal size band, one cut design, six development batches, four validation batches, and a limited number of fixture units. These constraints limit generalization. Abrasive retention on the scaife may differ from the feed distribution, and orientation, stock removal, temperature, and operator decisions may affect results. Changing stiffness may also change damping, mass, or contact behavior.

Additional facilities, operators, material lots, size bands, and longer production histories are needed to assess transferability and long-term process stability. Residual confounding may remain despite blocked randomization and recorded covariates.

The conformity limits have no demonstrated mapping to perceived beauty or commercial cut grades. The study does not establish color, clarity, crystal purity, durability, complete optical performance, long-term failure rates, or company-wide capability. Those questions require broader validated protocols and independent evidence.

7 Conclusions

This study evaluated abrasive upper-tail characteristics and fixture stiffness as interacting determinants of final-polishing accuracy in round-brilliant laboratory-grown diamonds. The split-plot design retained the conditioning episode as the abrasive experimental unit and separated stone-level outcomes from repeated facet measurements. A new-batch validation stage tested transfer beyond the development sample.

The restricted-tail abrasive with higher fixture stiffness produced a development mean RMSE of 0.044 degrees and an abrasive-dependent stiffness-benefit contrast of 0.024 degrees. In the held-out cohort, the selected N/H protocol reduced RMSE by a paired mean of 0.105 degrees, and all 24 stones met the prespecified first-pass criteria. Within the tested process, abrasive upper-tail control and fixture stiffness jointly improved facet accuracy and batch consistency.

Declarations

Author information and affiliation

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Support and competing interests

Daily Wear Diamonds provided the facilities, equipment, laboratory-grown diamond specimens, materials, and other resources required for the study. All authors are employees of Daily Wear Diamonds. This employment and institutional support constitute interests relevant to the work and are disclosed accordingly. The authors participated in the study and approved the decision to publish.

Contributions and approvals

All authors contributed to the conception or design of the study, acquisition or interpretation of data, and critical review of the manuscript. All authors approved the final manuscript and accept responsibility for the integrity of the work. Laboratory activities were conducted under the facility's applicable safety and equipment-qualification procedures.

Data availability and AI assistance

The study records supporting the reported tables and figures are available from the corresponding author on reasonable request, subject to protection of confidential commercial and material-identification information.

AI-assisted language editing was reviewed by the authors, who verified the methods, results, citations, and final text and take responsibility for the manuscript.

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